26900051 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCBTS301 - MOLECULAR BIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. The regulatory protein that binds to the operator of the lac operon is the: [U] / [CO2]

   a). Repressor b). Activator

   c). Polymerase d). Enhancer

2. Which RNA carries amino acids to the ribosome during protein synthesis? [K] / [CO1]

   a). mRNA b). rRNA

   c). tRNA d). miRNA

3. The central dogma describes the flow of information from: [K] / [CO1]

   a). Protein → RNA → DNA b). RNA → DNA → Protein

c). DNA → RNA → Protein d). Protein → DNA → RNA

4. RISC stands for: [K] / [CO4]

a). RNA-induced silencing complex b). RNA-induced synthesis complex

c). Ribosomal initiation sequence complex d). RNA intracellular synthesis complex

5. The process by which the ribosome moves along mRNA is called: [K] / [CO1]

a). Transcription b). Translocation

c). Replication d). Splicing

6. The promoter is a DNA sequence that primarily functions as: [K] / [CO2]

a). A site for DNA replication b). A site for initiation of transcription

c). A site for protein degradation d). A site for translation

7. The 5′ cap of eukaryotic mRNA contains: [U] / [CO2]

a). Adenine b). Guanine

c). 7-methylguanosine d). Uracil

8. DNA polymerase synthesizes the new DNA strand in which direction? [K] / [CO1]

a). 3′ → 5′ b). 5′ → 3′

c). Both directions d). Randomly

9. Which is a stop codon? [K] / [CO1]

a). UAA b). UGG

c). UAA d). GGG

10. Which radioactive isotope was used to label protein? [K] / [CO1]

a). ^{14}C b). ^{32}P

c). ^{35}S d). ^{15}N

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

11. What is the role of topoisomerase? [K] / [CO1]

12. Define transcription. [K] / [CO1]

13. What is attenuation in the trp operon? [U] / [CO2]

14. Define tRNA. [K] / [CO1]

15. List the major stages of translation. [K] / [CO1]

16. What is photoreactivation? [K] / [CO1]

17. What was the contribution of Erwin Chargaff to DNA structure? [K] / [CO1]

18. What is a regulator gene? Give one example [K] / [CO2]

19. What is conservative DNA replication? [U] / [CO1]

20. Why are aminoacyl-tRNA synthetases called “second genetic code readers”? [U] / [CO2]

21. What is siRNA? [K] / [CO4]

22. What is gene therapy? [U] / [CO5]

[2x10 = 20]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

23. Differentiate between cistron, muton and recon. [U] / [CO1]

24. Describe the structure of DNA. [U] / [CO1]

25. Discuss the ethical, legal and social implications of the Human Genome Project. [U] / [CO5]

26. Differentiate between theta, rolling circle and D-loop replication. [U] / [CO1]

27. Describe the different methods of post translation modification of protein. [U] / [CO2]

28. Explain the A, P and E sites of ribosomes. [U] / [CO1]

29. Describe the major levels at which eukaryotic gene expression is regulated. [U] / [CO2]

[4x5 = 20]